

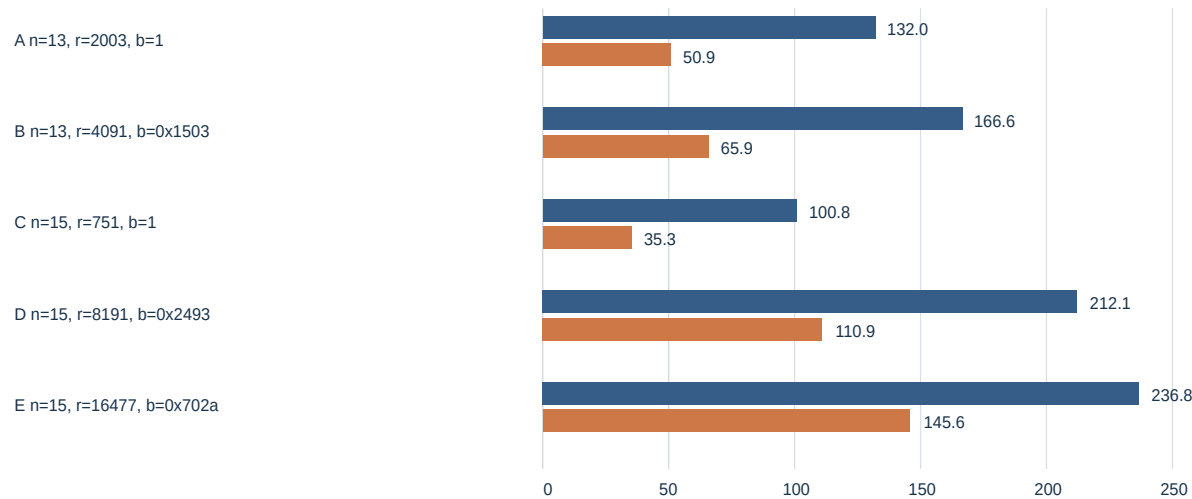
Exact binary-curve ECDLP comparison

9 October 2026 · Source: [research/ghs_curve_comparison_20261009/RESULT.md](#)

Question. How do ECDLP costs and characteristic-two GHS structure vary across exact binary-curve models, including models over the same field? The executable methods here are generic Pollard rho and BSGS. The higher-genus GHS solver is not implemented, so this report assigns it no ECDLP cost.

Counted ECDLP work differs across exact curve models

Mean group-addition equivalents per verified target; 8 targets per curve, one round, macOS L0



A-E map to full ICV1 identities in costs.tsv. Bars are sample means; no wall-time comparison or scaling inference.
Hash-table and canonicalisation work is counted separately but unpriced in this operation unit.

Measured one-target ECDLP work

Five registered models; eight independent targets each; one cold run per method and target. The native `ecbench` session `ECBS1hd395a1344225` verified all 80 scalars and exact replay reproduced all 80 records. Unit: $S = \text{counted group-addition equivalents} / \sqrt{r}$. Every figure is a counted lower bound because some native work, including hash-table work, is unpriced. The host earned L0; wall time is excluded.

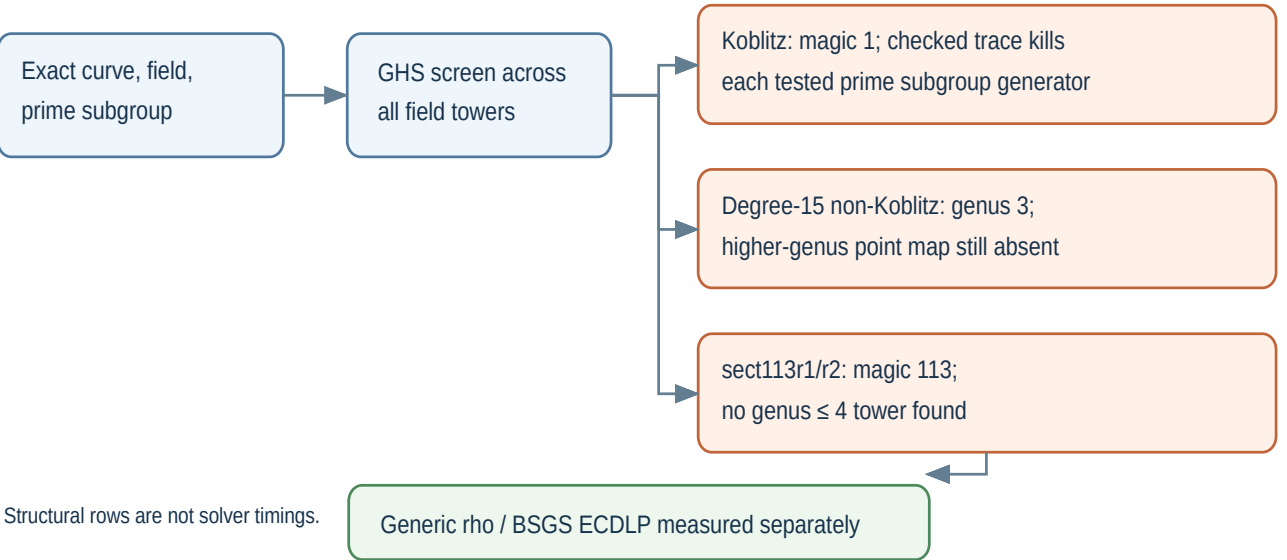
Curve ID suffix	Prime r	ρ .negation mean S	bsgs.negation mean S	BSGS / ρ	BSGS entries
t181-515ee569	2,003	2.949	1.137	0.385	24
t11-f036dc01	4,091	2.605	1.030	0.395	33
tm275-2d22ff5d	751	3.676	1.286	0.350	15
t5-d9fb183e	8,191	2.344	1.225	0.523	47
tm185-565c1669	16,477	1.844	1.134	0.615	66

Full IDs are `icv1-f2m13-t181-515ee569`, `icv1-f2m13-t11-f036dc01`, `icv1-f2m15-tm275-2d22ff5d`, `icv1-f2m15-t5-d9fb183e`, and `icv1-f2m15-tm185-565c1669`, in row order. The normalized ratios are paired on the same target. BSGS is a memory tradeoff; no physical speed ratio is claimed.

At degree 15, the mean absolute ρ count rises from 100.75 on the $r=751$ model to 212.125 and 236.75 on the $r=8,191$ and $r=16,477$ models. Mean BSGS counts are 35.25, 110.875, and 145.625. These are descriptive eight-target, one-round observations, not a scaling fit.

GHS screening and checked trace

What the curve comparison can actually test



Model	Best magic / genus	Checked trace outcome	Current status
13-bit Koblitz control	1 / 0	Generator killed	No subgroup transfer by trace
13-bit non-Koblitz	13 / 4095	Curve not subfield-defined	No genus ≤ 4 row
15-bit Koblitz control	1 / 0	Generator killed	No subgroup transfer by trace
15-bit non-Koblitz, b=0x2493	3 / 3	Curve not subfield-defined	Higher-genus model/map pending
15-bit non-Koblitz, b=0x702a	3 / 3	Curve not subfield-defined	Higher-genus model/map pending

A low GHS genus is a structural diagnostic. The Koblitz traces kill the tested prime subgroup generators; the genus-three cases require a smooth Jacobian model, point map and solver before any ECDLP cost can be measured. The separately frozen four-size Koblitz panel had 107 verified and 13 exhausted records; all exhaustions belonged to the signed-Frobenius rho arm. Its audit passed with 107 deterministic verified replays. The incomplete arm is retained as a diagnostic.

Standard and wide-field context

Exact checks of SEC 2's `sect113r1` and `sect113r2` give magic 113 and genus $2^{112}-1$ for their only proper tower, with no genus-four candidate. Neither curve model meets the checked trace precondition over F_2 . Their subgroup orders give a derived generic square-root scale near 2^{56} operations; no ECDLP solver was run at that size. The earlier binary F_2^{192} fixture transports an order-two point. NIST P-192 is a prime-field curve with derived generic scale near 2^{96} operations and is outside this binary GHS screen.

Evidence and remaining gates

Session ECBS1hd395a1 344 225: audit SHA-256 22cf241 164e32af826 128fdcf2e0ad1c770 768b071c71d54d34e52 409b67b973. Earlier panel ECBS1h5a490d269b02: audit SHA-256 bff5841f0732c0289aedc8fb37a6a89ab6d35a5662b0ef6cd7bfaa7b75aaa071. The Markdown report links the sealed sessions, per-curve intervals, commands, screen JSON, trace JSON, protocols, host and source hashes.

Next construction gate: for a specific genus-three model, construct and verify the smooth Jacobian, point transfer and subgroup order, then measure a complete index-calculus pipeline against rho on the same target. Until that gate passes, the ECDLP difference demonstrated here is the measured generic-method difference across exact subgroups.